VirtualSign: Revolutionizing Accessibility and Inclusivity in Education Through Automatic Sign Language Translation

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Abstract Promoting accessibility in education is crucial to ensure equal opportunities for all students. Assistive technologies, including an automatic bidirectional translator between sign and written languages, play a significant role in incentivizing inclusiveness and improving communication between the deaf and non-deaf communities. The translator utilizes advanced devices like the Microsoft Kinect and 5DT Sensor Gloves to capture hand motion and shape data, enabling accurate interpretation of sign language. Through machine learning and natural language processing, the translator processes and translates the data in real time. Known as VirtualSign, this technology promotes accessibility, social inclusion, and educational integration by facilitating communication and access to information while raising awareness of sign language in schools. Embracing these advancements supports equal access to education and empowers individuals with disabilities.

Keywords Accessibility in education · Equal opportunities · Assistive technologies · Automatic sign language translator

1 Introduction

In today's diverse world, effective communication plays a crucial role in fostering inclusivity and ensuring equal opportunities for all. However, individuals with hearing impairments often face significant barriers that hinder their communication experiences. These barriers can obstruct their educational journey and limit their ability to fully engage and participate in the learning process [1].

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Recognizing these challenges, the VirtualSign project emerges as a groundbreaking initiative that aims to enhance communication accessibility for deaf students in educational settings. Through the utilization of advanced technology and machine learning, VirtualSign revolutionizes the way individuals can access information in different languages [2–5].

The primary objective of the VirtualSign project is to create an inclusive environment in the field of education by breaking down linguistic barriers and fostering inclusivity. By focusing on the educational sphere, VirtualSign strives to provide deaf students with equal opportunities to engage and enhance learning [6].

VirtualSign offers a comprehensive suite of tools for assisted communication, empowering deaf students to overcome communication obstacles and actively participate in educational activities. These tools leverage state-of-the-art technology to bridge the gap between different languages and facilitate smooth and efficient communication.

By harnessing the power of technology, VirtualSign demonstrates the transformative potential of inclusive education. The project represents a significant milestone in leveraging technology to foster an environment where every individual, regardless of hearing impairments, can fully participate and benefit from educational experiences.

Moreover, the following paper explores (1) the theoretical framework and background of this project, focusing on the principles of sign language, the concept of inclusive education and the potential of assistive technologies, (2) the methodology of the project (i.e., concept and objectives), (3) the VirtualSign architecture and its technological foundations, and (4) the implementation and development of the VirtualSign technology in other areas and projects that demonstrate its versatility and potential impact beyond educational settings. It highlights real case studies where VirtualSign has been integrated into. These practical applications demonstrate the adaptability and scalability of VirtualSign in addressing communication barriers and promoting inclusivity across various contexts.

2 Theoretical Framework

Sign languages, including Portuguese Sign Language (LGP), exhibit distinct characteristics that distinguish them from spoken languages. They can be classified as either oral-auditory or visual-spatial languages, depending on the mode of communication used by the deaf community. Portuguese Sign Language, like other sign languages, possesses its own unique grammar and structure, allowing deaf individuals in Portugal to communicate effectively within their community [3].

However, despite the increasing interest and recognition of Portuguese Sign Language, there remains a significant challenge in providing adequate support for the deaf community to access written information in sign language. This limitation leads to educational information exclusion for disabled individuals who rely on sign language as their primary means of communication, which must be overcome by inclusive education.

Inclusive education is an approach to education that aims to provide equal opportunities and support for all learners, regardless [6, 7]. Its core principle is the recognition that every student deserves to be included in mainstream educational environments and actively participate in learning alongside their peers [8]. Implementing inclusive education can pose challenges, such as the need for adequate resources, specialized support services, and teacher training. However, by embracing inclusive education, educational systems can create environments that foster equality, and empower all students to thrive and enhance their academic growth and achievement [7]. One such resource is assistive technologies.

Assistive technologies have a profound impact on the learning experience of deaf students, providing numerous advantages. These technologies offer access to information, enhance communication, support language development, and provide academic assistance, all while fostering inclusive learning environments.

One notable advantage of technology is its automatic captioning and transcription services, which convert spoken language into written text. This feature allows deaf students to follow lectures, presentations, and multimedia content, improving their reading and comprehension skills and ensuring crucial information is accessible to them.

Furthermore, technology offers a range of visual learning tools that cater to diverse learning preferences and enhance conceptual understanding. These tools include multimedia presentations, educational videos with captions or sign language interpretation, interactive simulations, and visual aids. By providing multiple modalities for accessing information, technology empowers deaf students to grasp complex subjects effectively.

Online learning platforms also contribute to the educational journey of deaf students by offering flexible and accessible education. These platforms provide a wide range of educational materials, interactive exercises, and assessments, enabling deaf students to learn at their own pace, review content as needed, and access resources anytime, anywhere. Additionally, discussion forums and collaboration features foster peer interaction and support, creating a sense of community within the online learning environment.

By embracing these assistive technologies, educational institutions can foster the educational and personal growth of deaf students, ensuring equal opportunities for success. The integration of technology breaks down barriers, creating inclusive learning environments where deaf students can fully participate, engage, and thrive in their academic pursuits.

3 Methodology

3.1 Concept

The VirtualSign project overall goal was the creation of an assistive tool, an automatic bidirectional translator, available in classrooms, with the purpose to assist deaf students in educational settings in the access to information and communication. Individuals with such impairments often face significant challenges when it comes to communication and information. By recognizing these barriers, the VirtualSign project has emerged as a groundbreaking initiative aiming to bridge linguistic barriers and improve communication accessibility.

The VirtualSign project focuses on leveraging advanced technology to enhance accessibility for individuals with hearing impairments. By providing real-time sign language interpretation and communication support, VirtualSign enables individuals to effectively engage, and access information.

The VirtualSign project focuses on enhancing access to mainstream education for deaf students by utilizing VirtualSign Technology. Traditional educational methods often struggle to meet the unique needs and communication requirements of deaf students. By integrating sign language as an instructional language, the project aims to transform the educational experience, making educational materials more accessible and empowering deaf students to actively participate and excel in their studies.

By promoting the recognition and usage of sign language, the VirtualSign project fosters a diverse and inclusive environment that values the linguistic and cultural diversity of individuals with hearing impairments.

Through its multifaceted approach, the VirtualSign project strives to promote social inclusion, dismantle communication barriers, and create equal opportunities for individuals with hearing impairments.

3.2 Objectives

In today's society, promoting social inclusion and equal opportunities for people with disabilities, particularly those with hearing impairments, has become a key concern. Effective communication and access to information are often hindered by linguistic barriers faced by individuals with hearing impairments. Recognizing this challenge, the VirtualSign project has emerged as a groundbreaking initiative aimed at bridging these barriers and improving communication accessibility. With a focus on promoting social inclusion, enhancing communication, and addressing the needs of deaf students in education settings, the VirtualSign project strives to create an environment where individuals with hearing impairments have equal access, opportunities, and the ability to actively participate in society. To achieve this, the following objectives are pursued:

- Enhance communication accessibility: The VirtualSign project aims to break linguistic barriers and enhance communication accessibility for individuals with speech, hearing, and visual impairments. By leveraging the VirtualSign Technology, the project enables real-time sign language interpretation, empowering individuals to engage in effective communication.
- Promote social inclusion and equal opportunities: The project seeks to promote social inclusion by providing equal opportunities for people with disabilities, particularly those with hearing impairments. Through the implementation of VirtualSign Technology, the project aims to create an inclusive environment where individuals can actively participate in social interactions.
- Enhance educational experiences for deaf students: The VirtualSign project recognizes the challenges faced by deaf students in educational settings and aims to enhance their educational experiences. By providing innovative tools and platforms that integrate sign language as an instructional language, the project fosters effective communication between deaf students, teachers, and peers, empowering them to actively participate and excel in their studies.
- Promote recognition and usage of sign language: The VirtualSign project aims to promote the recognition and usage of sign language, acknowledging its importance as a means of communication. By embracing sign language as an integral part of society, the project strives to foster a more inclusive and diverse environment that values the linguistic and cultural diversity of individuals with hearing impairments.

Additionally, the project pursues the following specific objectives to support the implementation of the above goals:

- Develop and implement an accessible and user-friendly virtual platform: The VirtualSign project aims to create a virtual platform specifically designed to cater to the needs of deaf students. This platform will provide seamless access to educational resources, communication tools, and interactive learning experiences.
- Enhance communication capabilities: The project seeks to improve communication for deaf students by incorporating innovative technologies such as sign language recognition, real-time captioning, and video conferencing features. These tools will facilitate effective communication between students, teachers, and peers.
- Provide comprehensive educational content: The VirtualSign project aims to develop a diverse range of educational content that is specifically tailored to the needs of deaf students. This content will cover various subjects and educational levels, ensuring that students have equal access to quality education.
- Foster collaboration and social interaction: The project recognizes the importance of social interaction and collaboration in the learning process. It aims to create a virtual environment that encourages deaf students to actively engage with their peers, teachers, and the broader community, fostering a sense of inclusivity.

- Empower educators and support professionals: The VirtualSign project will provide training and resources to educators and support professionals to effectively utilize the virtual translator. This will ensure that teachers are equipped with the necessary skills and knowledge to create an inclusive learning environment.
- Conduct research and development: The project will actively engage in research and development to continuously improve the virtual platform, communication technologies, and educational content. By staying at the forefront of advancements in technology and education, the VirtualSign project can adapt and evolve to meet the evolving needs of deaf students.
- Advocate for policy changes and awareness: The VirtualSign project will actively advocate for policy changes at educational institutions and government levels to promote the inclusion of deaf students. Additionally, it will raise awareness about the challenges faced by deaf individuals and the importance of equal access to education and society.

By pursuing these objectives, the VirtualSign project aims to empower deaf students, bridge communication gaps, and create a more inclusive and equitable educational environment for individuals with hearing impairments.

3.3 Sustainability and Development

This section serves as a comprehensive roadmap for sustaining the impact and continuous development of the VirtualSign project. To ensure its long-term success, the project has strategically outlined various approaches. These include seeking sustainable funding through grants and partnerships committed to accessibility and inclusion, collaborating with global educational institutions and stakeholders, and actively engaging the community of users for ongoing feedback and input.

Additionally, the VirtualSign project prioritizes research and development, allocating resources to stay at the forefront of technology and educational methodologies. The scalability and adaptability of the VirtualSign platform are key considerations, ensuring it can evolve with emerging technologies and changing educational environments.

Through the implementation of these strategies, the VirtualSign project aims to guarantee the sustainability of its platform and contribute to the long-term success of deaf students in mainstream education.

4 VirtualSign Architecture

4.1 VirtualSign Model

VirtualSign aims to address the lack of adaptability in educational organizations for deaf and hard of hearing individuals, ensuring they have equal educational opportunities. The project includes the development of a model to integrate them into mainstream education, an animated virtual character for translating text-to-sign language, and a virtual reality environment for translating Portuguese Sign Language to Portuguese text. The model can be applied in various fields like virtual museums, web pages, and services. The project was initiated by an engineering school to assist in translating educational content into Portuguese Sign Language for undergraduate computer science degrees.

4.2 Automatic Sign Language Recognition

Sign language communication encompasses a combination of two-handed gestures and non-manual signals. However, recognizing sign language presents inherent challenges due to the diverse variations in signs, which can range from global body movements to intricate finger configurations. Each sign requires the acquisition of specific information, with varying levels of detail [9]. Therefore, striking a balance between capturing overall motion and preserving subtle local movements becomes crucial. Additionally, effectively handling occlusions is essential in this context.

The process of sign language recognition involves multiple steps, including hand, face, and body part tracking, feature extraction, modeling, analysis of time-varying signals, and integration of information from different modalities. To facilitate these tasks, a variety of sensors are employed to capture a wide range of features.

Use of Depth sensors. The introduction of the Microsoft Kinect depth sensor revolutionized technology by enabling computers to "see." Depth images obtained from Kinect are easier for computers to interpret compared to conventional color images. Processing depth information allows for more accurate object recognition, irrespective of color. Depth sensors have proven to be a robust tracking method, and the Kinect SDK provides accurate position and orientation data for hands [10].

Depth-based Hand Localization and Tracking. To capture hand gesture data, direct measures from a data glove with 14 sensors, such as the 5DT Data Glove 5 Ultra, were initially used. The 5DT Data Glove SDK already offered basic gesture recognition for 15 hand gestures. However, an updated version, the 5DT Ultra 17 gloves, includes additional gyroscope and accelerometer components situated in a small black box atop the glove. These additions enable the determination of hand rotation, which was not possible with the original model. Also, the IFG data gloves, developed in collaboration with the Instituto Federal de Goiás in Brazil, represent a newer model

being tested. Unlike the flexion sensor-based gloves, the IFG gloves solely rely on nine sensors, each containing an accelerometer and gyroscope. These sensors are strategically positioned on the fingers, wrist, and bottom of the index and ring fingers. While the IFG gloves offer advantages such as customization potential based on feedback, affordability, and the potential for a Kinect-free version, they require additional preprocessing to accurately predict hand configurations due to the absence of flexion sensors.

Building the Avatar and Hand-Coding Gestures. The aim was to develop a bidirectional translator for Portuguese Sign Language. This required recognizing signs based on body motion, translating them to written Portuguese, and simulating body motion representing written Portuguese. An avatar was built using a hand-coding approach that defines hand shape, orientation, location, and movement type for each sign in the vocabulary. Sign labels were classified by comparing recognized components with the ideal lexicon categories.

Analysis of Non-manual Signals. Facial recognition software like FaceReader typically focuses on basic emotions, while software like Faceshift uses depth sensors to accurately extract facial features and animate an avatar. Other non-manual signals, such as torso motion, can also be recognized using depth sensors. New gesture recognition devices, such as Omek Gasp and Leap Motion, provide accurate hand and finger tracking.

4.3 Text-to-Sign Translation and VSSO Configurator

The TTS translator and VSSO are online tools developed using Unity and WebGL technologies, ensuring compatibility across different web browsers. Users can conveniently access these tools from any location. The TTS translator enables users to input text and instantly view the corresponding sign language gestures performed by a 3D avatar. Currently, the project supports different European sign languages, offering a broad range of accessibility.

The VSSO configurator serves as a tool for building the sign language lexicon used in the project. Deaf individuals and sign language interpreters actively contribute to this process by adding and validating words through an intuitive interface reminiscent of a 3D animation tool. The configurator allows manipulation of the avatar's body, head, arms, facial expressions, and hand configurations. Expert validators, well-versed in sign language, ensure the accuracy of the signs created by editors. Additionally, the configurator provides a context feature to address any ambiguities that may arise during translation, further enhancing the quality and precision of the translations.

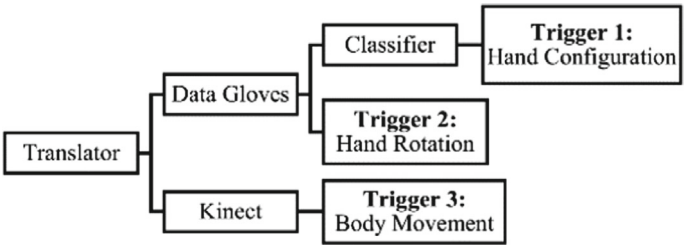


Fig. 1 Sign-to-text translator structure and triggers

4.4 Sign-to-Text Translation

The sign-to-text translator requires the use of data gloves and a Kinect sensor. A specialized hand-capture application is employed to generate datasets for classification, while the sign-to-text translator application performs the actual translation process. The system utilizes the data gloves, equipped with flexion sensors and a gyroscope, to capture hand movements, along with the Microsoft Kinect sensor for detecting both hand and body movements.

The sign-to-text translation tool generates a translation graph based on the established lexicon. Triggers, such as changes in hand configuration, hand rotation, and body movement, are employed to match gestures to words in real-time translation. The tool continuously monitors user input from the data gloves and Kinect sensor, activating the appropriate triggers to ensure accurate and timely translations (Fig. 1).

4.5 PowerPoint Add-On

To enhance accessibility and inclusivity in educational presentations, a PowerPoint add-on has been developed specifically for text-to-sign translation. This add-on empowers users to select text directly from a slide, choose the desired sign language, and receive a video presentation showcasing the 3D avatar performing the translation in real time. This feature greatly benefits teachers and students by seamlessly integrating sign language accessibility into their presentations, promoting effective communication for individuals with hearing impairments.

5 Enhancing the VirtualSign Technology: Revolutionizing Communication and Accessibility

The VirtualSign technology has emerged as a powerful tool to cater to the specific needs of deaf individuals in various projects, recognizing the challenges faced by deaf individuals in educational settings and public spaces. These initiatives have focused on leveraging this technology to promote inclusion and accessibility in education, cultural tourism, and technical knowledge dissemination. By integrating sign language automatic translation technology, these projects aim to bridge the communication gap between deaf and non-deaf individuals, provide equal learning opportunities, and enhance the overall experiences of deaf individuals in different domains.

Various projects have harnessed the VirtualSign technology to address the needs of deaf individuals in different contexts. In one project, a pedagogical model was proposed to deliver inclusive educational materials for the deaf through Massive Online Open Courses (MOOCs) integrated with sign language automatic translation technology. This model was implemented in a course on digital literacy for schoolteachers, which underwent evaluation to assess its inclusiveness and value for deaf students. The positive results encouraged the authors to extend the pedagogical model to create multilingual MOOCs, including Portuguese, Brazilian Portuguese, LGP, and Libras sign languages. The VirtualSign technology played a crucial role in providing sign language accessibility through a 3D avatar, enabling translation from spoken language to sign language and vice versa. Additionally, an audio add-on was included to benefit blind users [11].

The Inclusive Cultural Heritage Tourism (ICHT) project, on the other hand, is geared toward inclusive cultural tourism and addresses the needs of deaf tourists visiting cultural heritage spots. The system integrates an online platform and an automatic bidirectional sign language translator (ATT) for touristic activities. The ATT translator, based on the VirtualSign technology, enables communication between deaf and non-deaf users in various sign languages and spoken languages. The ICHT system also includes an informatics system installed on-site at cultural heritage destinations, providing information, ludic activities, and bilingual interfaces (avatar and hologram) in different languages [12–15].

Recognizing the specific challenges faced by deaf students in education, the TechWhiz project sought to enhance their learning experience. The TechWhiz platform focuses on providing equal access to education for deaf students by developing a semiautomatic glossary/dictionary in sign language. This glossary aims to explain technical and scientific concepts to deaf students, filling the gap that often arises when sign language interpreters are not available. The TechWhiz platform utilizes artificial intelligence, web scraping, and 3D animation technologies to offer explanations in sign language for technical terms and concepts [16].

The Assisted Communication for Education (ACE) and the International Assisted Communication for Education (I-ACE) projects aim to promote access to education and citizenship for individuals who are deaf or blind [1, 17, 18].

The ACE project focuses on enabling collaborations between different user types and interactions through various architectures such as collaborative art, serious games, and virtual classrooms. It acknowledges the creativity of blind individuals and seeks to leverage it through digital games and tools. Serious games have been developed for both deaf and blind users, promoting cooperation and communication [1, 17].

The I-ACE project specifically addresses the needs of deaf students in education and citizenship. It recognizes that access to education is severely compromised for the deaf and hearing impaired. The I-ACE project aims to establish an innovative infrastructure that supports live assisted communication between conventional classrooms and deaf students, as well as between deaf and non-deaf individuals. By bridging this communication gap, the project strives to improve integration, education, and overall quality of life for deaf students [18].

Moreover, the ACE and I-ACE projects emphasize the importance of increasing digital literacy among deaf and blind individuals, enhancing their communication competences through digital arts, and providing inclusive learning environments. By leveraging technology and innovative approaches, these projects aim to empower individuals with disabilities, enhance their learning experiences, and promote equal opportunities for education and participation in society.

All of these projects leveraged the VirtualSign technology developed by the Polytechnic Institute of Porto, which facilitated the translation between spoken and sign languages. The VirtualSign 3D avatar and its translation infrastructure served as the foundation for sign language accessibility in both the pedagogical MOOCs and the TechWhiz glossary. The ICHT project expanded on this technology to develop an innovative solution for inclusive tourism, using computer vision, machine translation, holography, and 3D animations to create a unique user interface for deaf and non-deaf users.

By harnessing the VirtualSign technology, these projects aimed to bridge the communication gap between deaf and non-deaf individuals, promote inclusion in education and tourism, and facilitate access to cultural heritage sites for the deaf.

6 Conclusions

The integration of technology plays a crucial role in empowering deaf students by granting access to information, fostering communication, and expanding educational resources. The VirtualSign system revolutionizes communication by enabling individuals with different languages to interact, bridging the gap between deaf and non-deaf individuals worldwide. Through machine learning and advanced technology like Kinect and 5dt Gloves, VirtualSign achieves high accuracy in recognizing sign language. With a strong focus on promoting social inclusion and meeting the needs of deaf students in educational settings, the project aims to create an environment where individuals with hearing impairments have equal access and opportunities. By

implementing VirtualSign in educational institutions, communication between deaf and non-deaf individuals can be greatly enhanced.

Future plans for VirtualSign in education involve developing a gloveless system with the same precision in gesture-to-text translation, making the technology more accessible in classrooms. The project also aims to create an intermediate semantic system that seamlessly bridges sign language and written text, facilitating communication between deaf and non-deaf individuals.

The VirtualSign system serves as the foundation for other innovative projects aimed at enhancing the educational experience of deaf students, but also among other domains, such as cultural tourism. By leveraging the success of VirtualSign, these projects build upon its principles of inclusivity and accessibility, and fosters communication.

Overall, the VirtualSign project is dedicated to creating an inclusive and accessible learning environment, enabling deaf students to fully engage, communicate, and express themselves in educational activities. By leveraging innovative technology, VirtualSign promotes equal opportunities and social inclusion in education. It represents a significant advancement in assisted communication, fostering inclusivity and enhancing the educational experience in any area of study for individuals with hearing impairments.

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